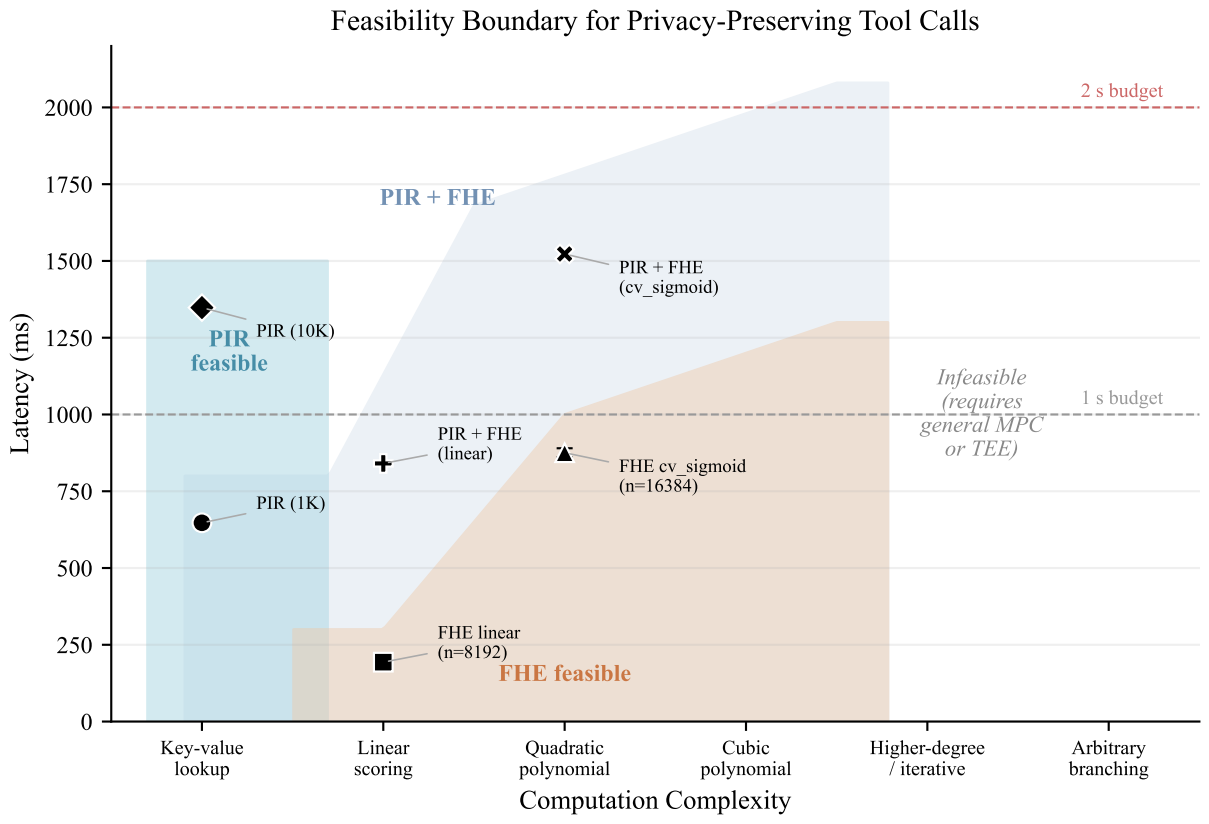
Feasibility Boundary for Privacy-Preserving Tool Calls
Latency (ms)
2000
1750
1500
1250
1000
750
500
250
0
2 s budget
1 s budget
PIR + FHE
PIR feasible
FHE feasible
PIR (10K)
PIR (1K)
PIR + FHE (linear)
PIR + FHE (cv_sigmoid)
FHE cv_sigmoid (n=16384)
FHE linear (n=8192)
Infeasible (requires general MPC or TEE)
Key-value lookup
Linear scoring
Quadratic polynomial
Cubic polynomial
Higher-degree / iterative
Arbitrary branching
Computation Complexity